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IMAGE FORMING APPARATUS

Abstract

A movable chute is located downstream of a transfer unit and upstream of a fuser in a first direction. A downstream end of the movable chute located at a second position is farther from a sheet conveyance path than the downstream end of the movable chute located at a first position. A reconveyance chute forms a reconveyance path of a sheet that is reconveyed from the fuser toward the transfer unit. The reconveyance path is located between the movable chute and the reconveyance chute in a sheet thickness direction. The movable chute has a sheet conveyance surface and a first reconveyance surface. The sheet conveyance surface guides the sheet that is conveyed from the transfer unit toward the fuser along the sheet conveyance path. The first reconveyance surface faces opposite to the sheet conveyance surface in the sheet thickness direction. The first reconveyance surface forms the reconveyance path.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority from Japanese Patent Application No. 2024-030273 filed on Feb. 29, 2024. The entire content of the priority application is incorporated herein by reference.

BACKGROUND ART

[0002] An image forming apparatus including a transfer unit that transfers a toner image onto a sheet and a fuser that fixes the toner image onto the sheet is known.

SUMMARY

[0003] For example, an image forming apparatus includes a guide member for guiding a sheet from a transfer unit to a fuser, and a reconveyance tray for guiding the sheet which is reversed after passing through the fuser and is conveyed toward the transfer unit.

[0004] In such an image forming apparatus, the guide member is configured to be movable, so that the sheet is efficiently conveyed in response to deflection (bending) generated in the sheet conveyed between the transfer unit and the fuser.

[0005] However, in the image forming apparatus including the guide member and the reconveyance tray, the reconveyance tray is located between the guide member and a reconveyance path of the sheet conveyed from the fuser to the transfer unit. Thus, in order to prevent the interference (contact) between the reconveyance tray and the guide member, the movement range of the guide member is restricted, and there is a possibility that the sheet cannot be efficiently conveyed.

[0006] In view of the foregoing, an example of an object of this disclosure is to provide an image forming apparatus configured to efficiently convey a sheet by preventing restriction of a movement range of a guide member.

[0007] According to one aspect, this specification discloses an image forming apparatus. The image forming apparatus includes a transfer unit, a fuser, a movable chute, and a reconveyance chute. The transfer unit is configured to transfer a toner image onto a sheet. The fuser includes a heater and is configured to fix the toner image on the sheet conveyed from the transfer unit. The movable chute is located downstream of the transfer unit and upstream of the fuser in a first direction. The first direction is a sheet conveyance direction from the transfer unit toward the fuser. The movable chute has a downstream end in the first direction. The movable chute is movable between a first position and a second position different from the first position. The downstream end of the movable chute located at the second position is farther from a sheet conveyance path than the downstream end of the movable chute located at the first position. The sheet conveyance path is formed along the first direction. The reconveyance chute is configured to guide the sheet having passed the fuser to the transfer unit. The reconveyance chute forms a reconveyance path of the sheet that is reconveyed from the fuser toward the transfer unit. The reconveyance path is located between the movable chute and the reconveyance chute in a sheet thickness direction of the sheet. The movable chute has a sheet conveyance surface and a first reconveyance surface. The sheet conveyance surface is configured to guide the sheet that is conveyed from the transfer unit toward the fuser along the sheet conveyance path. Thus, the sheet conveyed along the sheet conveyance path is guided by the sheet conveyance surface of the movable chute. The first reconveyance surface faces opposite to the sheet conveyance surface in the sheet thickness direction. The first reconveyance surface forms the reconveyance path. Thus, the sheet conveyed along the reconveyance path is guided by the first

reconveyance surface of the movable chute.

[0008] According to the present disclosure, the movement range of the movable chute is not restricted when the movable chute is moved to the second position. This secures a large space in which the sheet conveyance path is located, and enables efficient conveyance of the sheet. Further, the number of components constituting the image forming apparatus is reduced, and the image forming apparatus is downsized in the upper-lower direction.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] FIG. 1 is a schematic central cross-sectional view showing an image forming apparatus.

[0010] FIG. 2 is a perspective view of the image forming apparatus with a front cover in an open position.

[0011] FIG. 3 is a perspective view showing a first main body frame, a second main body frame, an air duct, and a movable chute, as viewed from left front.

[0012] FIG. 4 is a side cross-sectional view showing a fuser.

[0013] FIG. 5 is a perspective view showing a heating unit and a pressure roller of the fuser.

[0014] FIG. 6 is a central cross-sectional view showing the image forming apparatus.

[0015] FIG. 7 is a side cross-sectional view showing a region between the transfer unit and the fuser in an apparatus main body in a state where the movable chute is at a first position.

[0016] FIG. 8 is a side cross-sectional view showing the region between the transfer unit and the fuser in the apparatus main body in a state where the movable chute is at a second position.

[0017] FIG. 9 is a perspective view showing the movable chute to which a chute sheet metal is fixed.

[0018] FIG. 10 is an exploded perspective view showing the chute sheet metal, the movable chute, and a chute stopper.

[0019] FIG. 11A is a plan view showing the movable chute to which the chute sheet metal is fixed.

[0020] FIG. 11B is a bottom view showing the movable chute to which the chute sheet metal is fixed.

[0021] FIG. 12 is a side cross-sectional view showing the movable chute to which the chute sheet metal is fixed.

[0022] FIG. 13 is a perspective view showing a fuser chute, the movable chute, and a guide frame.

[0023] FIG. 14 is a bottom view showing the fuser chute, the movable chute, and the guide frame.

[0024] FIG. 15 is a cross-sectional view taken along a line XV-XV in FIG. 14.

[0025] FIG. 16 is a cross-sectional view taken along a line XVI-XVI in FIG. 14.

[0026] FIG. 17 is a block diagram showing a drive mechanism of the movable chute.

[0027] FIG. 18 is a rear cross-sectional view showing a chute link in a state of being spaced apart from the movable chute.

[0028] FIG. 19 is a rear cross-sectional view showing the chute link in a state of pressing the movable chute;

[0029] FIG. 20A is a perspective view showing a movable chute and a chute sheet metal.

[0030] FIG. 20B is a plan view showing the movable chute and the chute sheet metal.

[0031] FIG. 21 is a bottom view showing a fuser chute and the movable chute.

[0032] FIG. 22 is a cross-sectional view taken along a line XXII-XXII in FIG. 21.

[0033] FIG. 23 is a side cross-sectional view showing a movable chute in a state of being located at a first position.

[0034] FIG. 24 is a side cross-sectional view showing the movable chute in a state of being located at a second position.

DESCRIPTION

[0035] Next, embodiments for carrying out the present disclosure will be described with reference to the accompanying drawings.

[0036] An image forming apparatus **1** shown in FIG. **1** is an embodiment of an image forming apparatus including a sheet conveyor according to the present disclosure. The image forming apparatus **1** is a laser printer that forms an image on a sheet **S** by an electrophotographic method.

[0037] In the following description, the right side in FIG. **1** is defined as the front side of the image forming apparatus **1**, the left side in FIG. **1** is defined as the rear side of the image forming apparatus **1**, the near side in a direction perpendicular to the drawing sheet of FIG. **1** is defined as the left side of the image forming apparatus **1**, and the far side is defined as the right side of the image forming apparatus **1**. The upper side and the lower side in FIG. **1** are defined as the upper side and the lower side of the image forming apparatus **1**, respectively.

[0038] The image forming apparatus **1** includes an apparatus main body **2**, a sheet feed unit **3**, an image forming unit (print engine) **5**, a fuser **6**, and a sheet discharge unit **9**.

[0039] The apparatus main body **2** accommodates the sheet feed unit **3**, the image forming unit **5**, the fuser **6**, and the sheet discharge unit **9**. An opening **2A** is formed in a front surface of the apparatus main body **2**, and the apparatus main body **2** includes a front cover **21** configured to open and close the opening **2A**. The front cover **21** is configured to be pivotable about a pivot shaft **21a** at a lower end thereof. By pivoting about the pivot shaft **21a**, the front cover **21** is movable between a closed position at which the front cover **21** closes the opening **2A** and an open position at which the front cover **21** opens the opening **2A**.

[0040] The sheet feed unit **3** includes a sheet feed tray **10** for supporting the sheet **S**, a sheet conveyance unit **30**, a conveyance roller pair **34**, and a registration roller **35a**. The sheet feed unit **3** is disposed in a lower part of the apparatus main body **2** and conveys the sheet **S** supported by the sheet feed tray **10** to the image forming unit **5**. The image forming apparatus **1** has a conveyance path **P1** for the sheet **S** from the sheet feed unit **3** to the sheet discharge unit **9** via the image forming unit **5**. The conveyance path **P1** is an example of a sheet conveyance path.

[0041] The sheet feed tray **10** includes a pressure plate **12** and a pressing plate **13**. The pressure plate **12** is a plate-like member that supports the sheet **S** from below. The pressure plate **12** is pivotable about a pivot point **12a** at a rear end, and is movable up and down between a lowered position and a raised position by pivoting about the pivot point **12a**. The pressing plate **13** is located below the pressure plate **12** and moves the pressure plate **12** up and down between the lowered position and the raised position.

[0042] The sheet conveyance unit **30** is a conveyance mechanism that separates and takes out the sheets **S** supported by the sheet feed tray **10** one sheet at a time and conveys the sheets **S** toward the image forming unit **5**. The sheet conveyance unit **30** includes a sheet feed roller **31**, a separation roller **32**, and a separation pad **33**.

[0043] The sheet feed roller **31** is a roller for feeding the sheet **S** supported by the sheet feed tray **10** toward the separation roller **32**. The separation roller **32** is disposed downstream of the sheet feed roller **31** in a sheet conveyance direction. The separation pad **33** is disposed to face the separation roller **32** and is urged toward the separation roller **32**.

[0044] The sheets **S** fed toward the separation roller **32** by the sheet feed roller **31** are separated one sheet at a time between the separation roller **32** and the separation pad **33**. The sheets **S** separated one sheet at a time are sent out to the conveyance path **P1**.

[0045] The sheet **S** sent to the conveyance path **P1** is conveyed toward the image forming unit **5** by the conveyance roller pair **34**, the registration roller **35a**, and a pinch roller **35b** disposed to face the registration roller **35a**. The registration roller **35a** regulates and temporarily stops movement of the leading edge of the sheet **S** conveyed from the sheet feed tray **10** toward the image forming unit **5**, corrects skew of the sheet **S**, and then conveys the sheet **S** toward the image forming unit **5** at a particular timing. The registration roller **35a** is a roller for correcting the skew of the sheet **S** conveyed toward a transfer unit **5A** included in the image forming unit **5**.

[0046] The image forming apparatus **1** includes a pre-registration sensor **97** located upstream of the registration roller **35a** in the sheet conveyance direction and a post-registration sensor **98** located downstream of the registration roller **35a** in the sheet conveyance direction.

[0047] The pre-registration sensor **97** is a sensor for detecting the sheet **S** conveyed along the conveyance path **P1**. The pre-registration sensor **97** is configured to be turned on upon detecting the leading edge of the sheet **S** when the leading edge of the sheet **S** passes the pre-registration sensor **97**, and to be turned off upon detecting the trailing edge of the sheet **S** when the trailing edge of the sheet **S** passes the pre-registration sensor **97**, for example.

[0048] The post-registration sensor **98** is a sensor for detecting the sheet **S** conveyed along the conveyance path **P1**. The post-registration sensor **98** is configured to be turned on upon detecting the leading edge of the sheet **S** when the leading edge of the sheet **S** passes the post-registration sensor **98**, and to be turned off upon detecting the trailing edge of the sheet **S** when the trailing edge of the sheet **S** passes the post-registration sensor **98**, for example.

[0049] The image forming unit **5** is disposed downstream of the sheet feed unit **3** in the sheet conveyance direction, and forms an image on the sheet **S** conveyed from the sheet feed unit **3**. The image forming unit **5** includes a process cartridge **50**, a transfer roller **55**, and an exposure unit **56**. The process cartridge **50** forms an image onto a surface of the sheet **S** conveyed from the sheet feed unit **3**. The transfer roller **55** is disposed to face a photosensitive drum **54** of the process cartridge **50**. The exposure unit **56** exposes a surface of the photosensitive drum **54**. The process cartridge **50** is an example of a cartridge that is attachable to and detachable from the apparatus main body.

[0050] The process cartridge **50** is disposed above the sheet feed unit **3** in the apparatus main body **2**. The process cartridge **50** includes a developer storage chamber **51**, a supply roller **52**, a development roller **53**, and the photosensitive drum **54**. The process cartridge **50** supports the pinch roller **35b**.

[0051] The process cartridge **50** includes a drum cartridge and a development cartridge. The drum cartridge includes the photosensitive drum **54**. The development cartridge is attached to the drum cartridge and includes the development roller **53**. The process cartridge **50** is detachably mounted on the apparatus main body **2**. In this case, the process cartridge **50** is attached to the apparatus main body **2** by inserting the drum cartridge and the development cartridge attached to the drum cartridge, as a unit, into the apparatus main body **2**. The process cartridge **50** is detached from the apparatus main body **2** by pulling the drum cartridge and the development cartridge attached to the drum cartridge, as a unit, out of the apparatus main body **2**.

[0052] The process cartridge **50** is an example of a cartridge, and may include at least one of a photosensitive drum, a development roller, and a toner container. The process cartridge **50** is attachable to and detachable from the apparatus main body **2** when the front cover **21** is in the open position.

[0053] The photosensitive drum **54** is disposed in a posture in which an axis **X** is along a left-right direction (sheet width direction) in a state where the process cartridge **50** is attached to the apparatus main body **2**. The photosensitive drum **54** includes a drum shaft **54a** made of metal and extending in an axial direction along the axis **X**. The photosensitive drum **54** rotates about the drum shaft **54a**.

[0054] In the present embodiment, the process cartridge **50** includes the drum cartridge having the photosensitive drum **54** and the development cartridge attached to the drum cartridge and having the development roller **53**. Alternatively, the process cartridge **50** may be configured to include a cartridge and a toner box. The cartridge includes the photosensitive drum **54** and the development roller **53**. The toner box is attached to the cartridge and contains toner.

[0055] Further, the image forming apparatus may be configured such that a drum cartridge having the photosensitive drum **54** and a development cartridge having the development roller **53** are separately attached to the apparatus main body **2**. That is, the process cartridge **50** may include at least one of a photosensitive drum, a development roller, and a toner container, and may be

configured to be attachable to and detachable from the apparatus main body.

[0056] The exposure unit **56** includes a laser diode, a polygon mirror, a lens, a reflecting mirror, and so on. The exposure unit **56** irradiates the photosensitive drum **54** with laser light based on image data input to the image forming apparatus **1**, thereby exposing the surface of the photosensitive drum **54**.

[0057] The developer storage chamber **51** stores toner serving as developer. The toner stored in the developer storage chamber **51** is sent to the supply roller **52** while being agitated by an agitating member (not shown). The supply roller **52** further supplies the toner sent from the developer storage chamber **51** to the development roller **53**.

[0058] The development roller **53** is disposed in close contact with the supply roller **52**, and bears the toner supplied from the supply roller **52** and positively charged by a sliding contact member (not shown). A developing bias is applied to the development roller **53** by a bias applying unit (not shown).

[0059] The photosensitive drum **54** is disposed adjacent to the development roller **53**. The surface of the photosensitive drum **54** is uniformly charged by a charger (not shown) and then exposed by the exposure unit **56**. The exposed portion of the photosensitive drum **54** has a lower potential than the other portion, and an electrostatic latent image based on image data is formed on the photosensitive drum **54**. Then, positively charged toner is supplied from the development roller **53** to the surface of the photosensitive drum **54** on which the electrostatic latent image is formed, whereby the electrostatic latent image is visualized to form a toner image.

[0060] The transfer roller **55** is disposed to face the photosensitive drum **54**, and a transfer bias is applied to the transfer roller **55** by a bias applying unit (not shown). In a state where the transfer bias is applied to the surface of the transfer roller **55**, the sheet **S** is nipped and conveyed between the photosensitive drum **54** on which the toner image is formed and the transfer roller **55**, whereby the toner image formed on the surface of the photosensitive drum **54** is transferred onto the surface of the sheet **S**.

[0061] The process cartridge **50**, the photosensitive drum **54**, and the transfer roller **55** constitute the transfer unit **5A** for transferring the toner image onto the sheet **S**. The sheet **S** to which the toner image has been transferred is conveyed from the transfer unit **5A** to the fuser **6**. The sheet conveyance direction from the transfer unit **5A** toward the fuser **6** along the conveyance path **P1** is defined as a first direction **D1**.

[0062] The fuser **6** includes a heating unit **61** and a pressure roller **62**, and fixes the image transferred to the sheet **S** by the transfer unit **5A**. The heating unit **61** is heated by supplying electric power from a power source (not shown). The heating unit **61** is an example of a heating unit. The pressure roller **62** is disposed to face the heating unit **61**. One of the heating unit **61** and the pressure roller **62** is urged toward the other by an urging mechanism (not shown), and the heating unit **61** and the pressure roller **62** are in close contact with each other.

[0063] When the sheet **S** to which the toner image has been transferred is conveyed to the fuser **6**, the sheet **S** is conveyed while being nipped between the heating unit **61** and the pressure roller **62**, and the sheet **S** is heated, and the toner image is fixed to the sheet **S**. In this way, the fuser **6** fixes the toner image to the sheet **S** conveyed from the transfer unit **5A**. The sheet **S** on which the toner image is fixed by the fuser **6** is conveyed to the sheet discharge unit **9** by a post-fixing conveyance roller pair **68** disposed downstream of the fuser **6** in the sheet conveyance direction.

[0064] The image forming apparatus **1** includes a sheet sensor **99** located between the fuser **6** and the post-fixing conveyance roller pair **68**. The sheet sensor **99** is a sensor for detecting the sheet **S** conveyed along the conveyance path **P1**. The sheet sensor **99** is configured to be turned on upon detecting the leading edge of the sheet **S** when the leading edge of the sheet **S** passes the sheet sensor **99**, and to be turned off upon detecting the trailing edge of the sheet **S** when the trailing edge of the sheet **S** passes the sheet sensor **99**, for example.

[0065] The sheet discharge unit **9** is located downstream of the post-fixing conveyance roller pair

68 in the sheet conveyance direction. The sheet discharge unit **9** discharges the sheet **S** on which the image is formed by the image forming unit **5** to the outside of the image forming apparatus **1**, or conveys the sheet **S** again toward the image forming unit **5**. The sheet discharge unit **9** includes a sheet discharge roller pair **291** and a sheet discharge tray **292**. The sheet discharge tray **292** is provided at the apparatus main body **2**. The printed sheet **S** on which an image is formed in the image forming unit **5** is discharged to the sheet discharge tray **292**.

[0066] The conveyance path **P1** on the downstream side of the fuser **6** in the conveyance path **P1** is formed as a discharge path **P1a**. The sheet discharge roller pair **291** is configured to discharge the sheet **S** conveyed from the fuser **6** along the discharge path **P1a** toward the outside of the apparatus main body **2**. The discharge path **P1a** is a path for guiding the printed sheet **S** to the discharge tray **292**. The sheet discharge tray **292** is formed on the upper surface of the apparatus main body **2** and supports the sheet **S** discharged to the outside of the apparatus main body **2** by the sheet discharge roller pair **291**.

[0067] The sheet discharge roller pair **291** is configured to be rotatable in a discharge direction which is a rotation direction for conveying the sheet **S** toward the sheet discharge tray **292** and a reconveyance direction which is a rotation direction opposite to the discharge direction. The image forming apparatus **1** includes a reconveyance path **P2** that guides the one-side printed sheet **S**, which has been conveyed along the conveyance path **P1** and passed the fuser **6**, to the conveyance path **P1** on the upstream side of the registration roller **35a** in the sheet conveyance direction.

[0068] The reconveyance path **P2** branches from the discharge path **P1a** at a branch point **Pb** located between the post-fixing conveyance roller pair **68** and the discharge roller pair **291**, extends forward between the image forming unit **5** and the sheet feed tray **10**, and joins the conveyance path **P1** at a merging point **Pa** located between the conveyance roller pair **34** and the registration roller **35a**.

[0069] The sheet **S** conveyed from the fuser **6** to the sheet discharge unit **9** is conveyed again to the image forming unit **5** through the reconveyance path **P2** by reversing the rotation direction of the sheet discharge roller pair **291** to rotate in the reconveyance direction. The sheet **S** conveyed to the reconveyance path **P2** by the sheet discharge roller pair **291** is conveyed toward the image forming unit **5** by a first reconveyance roller pair **36** and a second reconveyance roller pair **37** provided on the reconveyance path **P2**.

[0070] The sheet **S** conveyed along the reconveyance path **P2** is conveyed to the transfer unit **5A** of the image forming unit **5** after passing through the merging point **Pa**. Thus, the reconveyance path **P2** is a conveyance path of the sheet **S** conveyed from the fuser **6** toward the transfer unit **5A**. The sheet conveyance direction of the sheet **S** conveyed along the reconveyance path **P2** toward the transfer unit **5A** is a second direction **D2**.

[0071] The image forming apparatus **1** is configured to perform duplex printing in which the sheet **S** having an image formed on one side thereof by the image forming unit **5** is conveyed again to the image forming unit **5** through the reconveyance path **P2** and an image is formed on the other side of the sheet **S**.

[0072] As shown in FIGS. **2** and **3**, the apparatus main body **2** includes a first main body frame **26** and a second main body frame **27** which are disposed to be separated from each other in the left-right direction. The first main body frame **26** is located at a right end of the apparatus main body **2**, and the second main body frame **27** is located at a left end of the apparatus main body **2**. The first main body frame **26** extends in the front-rear direction and the upper-lower direction, and is disposed in a posture in which a plate surface faces in the left-right direction. The second main body frame **27** extends in the front-rear direction and the upper-lower direction, and is disposed in a posture in which a plate surface faces in the left-right direction.

[0073] The fuser **6** and the transfer unit **5A** including the process cartridge **50** are disposed between the first main body frame **26** and the second main body frame **27**. The first main body frame **26** is disposed on the right side of the transfer unit **5A** and the fuser **6**, and the second main body frame

27 is disposed on the left side of the transfer unit **5A** and the fuser **6**. The process cartridge **50** constituting the transfer unit **5A** is detachably supported by the first main body frame **26** and the second main body frame **27**.

[0074] The first main body frame **26** supports a fan **29** for discharging air in the apparatus main body **2** to the outside of the apparatus main body **2**. The second main body frame **27** supports a control board **951** on which electronic components such as a CPU **951a** are mounted. The control board **951** constitutes a controller **95** (see FIG. **17**) for controlling the operation of the image forming apparatus **1**.

[0075] An air duct **28** extending along the left-right direction is disposed between the first main body frame **26** and the second main body frame **27**. The air duct **28** is provided to span between the first main body frame **26** and the second main body frame **27**. The right end of the air duct **28** is supported by the first main body frame **26**, and the left end of the air duct **28** is supported by the second main body frame **27**.

[0076] As shown in FIG. **6**, the air duct **28** is located between the transfer unit **5A** and the fuser **6** in the front-rear direction. The air duct **28** is disposed at a position overlapping the fan **29** when viewed in the left-right direction. Air flows through the air duct **28**. The air duct **28** guides air in the apparatus main body **2** toward the fan **29**.

[0077] As shown in FIGS. **4** and **5**, the heating unit **61** of the fuser **6** includes a heater **611**, a holder **612**, a stay **613**, and a belt **614**. The heater **611** is a plate-like heater extending in the left-right direction. The heater **611** has a first surface **611A** and a second surface **611B** opposite to the first surface **611A**. The first surface **611A** is supported by the holder **612**.

[0078] The holder **612** is made of, for example, a plastic member, and has a guide surface **612a** and a support wall **612b**. The guide surface **612a** contacts an inner peripheral surface **614a** of the belt **614** to guide the belt **614**. The support wall **612b** has a support surface **612A** that supports the heater **611**. The support surface **612A** of the support wall **612b** contacts the first surface **611A** of the heater **611**. The stay **613** is a member that supports the holder **612**, and is formed by bending a plate material having a rigidity higher than that of the holder **612**, for example, a steel plate, into a substantially U-shape in cross-section.

[0079] The belt **614** is an endless belt having heat resistance and flexibility, and may include a metal tube made of metal such as stainless steel and a fluororesin layer covering the metal tube. The heater **611**, the holder **612**, and the stay **613** are disposed inside the belt **614**. The belt **614** is configured to rotate around the heater **611**, the holder **612**, and the stay **613**. The inner peripheral surface **614a** of the belt **614** contacts the heater **611**.

[0080] The pressure roller **62** includes a metal shaft **62A** and an elastic layer **62B** covering the shaft **62A**. The pressure roller **62** is pressed against the heater **611** via the belt **614**. The pressure roller **62** and the heater **611** sandwich the belt **614** therebetween, thereby forming a nip NP for heating and pressurizing the sheet S. That is, the pressure roller **62** forms the nip NP with the belt **614**, and heats and presses the sheet S together with the heater **611** at the nip NP.

[0081] The pressure roller **62** is driven to rotate by a driving force transmitted from a motor included in the image forming apparatus **1**. The pressure roller **62** is rotationally driven to cause the belt **614** to rotate by a frictional force with the sheet S nipped at the nip NP. Thus, the sheet S on which the toner image is transferred is conveyed between the pressure roller **62** and the heated belt **614**, and the toner image is thermally fixed.

[0082] As shown in FIGS. **1** and **6** to **8**, the fuser **6** includes a fuser frame **63** that covers the heating unit **61** and the pressure roller **62**. The fuser frame **63** holds the heating unit **61** and the pressure roller **62**. The fuser frame **63** covers the periphery of the heating unit **61** and the pressure roller **62**. The fuser frame **63** has a first opening **63a** formed upstream of the heating unit **61** and the pressure roller **62** in the first direction D1, and a second opening **63b** formed downstream of the heating unit **61** and the pressure roller **62** in the first direction D1.

[0083] The fuser frame **63** includes an upper frame **63A** and a lower frame **63B**. The upper frame

63A is located above the lower frame **63B**. The upper frame **63A** covers the heating unit **61**, and the lower frame **63B** covers the pressure roller **62**.

[0084] In the present embodiment, the fuser **6** includes the heating unit **61** having the heater **611** and the belt **614**, and the pressure roller **62**. Alternatively, the fuser **6** may include a heating roller having a heater therein and a pressure roller pressed against the heating roller. Alternatively, the fuser **6** may include a heating roller having a heater therein and a pressure belt pressed against the heating roller by an elastic member.

[0085] As shown in FIGS. **1** and **6**, the apparatus main body **2** includes a reconveyance chute **23** that forms the reconveyance path **P2**. The reconveyance chute **23** is disposed between the sheet feed tray **10** and the transfer unit **5A** in the upper-lower direction. The reconveyance chute **23** is located below the reconveyance path **P2**, and has a lower chute surface **23a** that forms a lower surface of the reconveyance path **P2**. The reconveyance chute **23** guides the sheet **S** that has passed the fuser **6** and has been reversed to the transfer unit **5A** along the reconveyance path **P2**.

[0086] The reconveyance chute **23** is detachably attached to the apparatus main body **2**. The reconveyance chute **23** is configured to be movable in the front-rear direction. The reconveyance chute **23** attached to the apparatus main body **2** is detached from the apparatus main body **2** by being pulled out rearward.

[0087] As shown in FIGS. **6** to **8**, the image forming apparatus **1** includes a fuser chute **24**. The fuser chute **24** is located upstream of the fuser **6** in the first direction **D1** and is supported by the lower frame **63B** of the fuser frame **63**. The fuser chute **24** is located below the conveyance path **P1** and forms a lower surface of the conveyance path **P1**. The fuser chute **24** has a sheet conveyance surface **24A** facing upward. The fuser chute **24** forms the lower surface of the conveyance path **P1** by the sheet conveyance surface **24A**. The sheet conveyance surface **24A** guides the lower side of the sheet **S** conveyed along the conveyance path **P1**.

[0088] The fuser chute **24** is located above the reconveyance path **P2** and forms an upper surface of the reconveyance path **P2**. The fuser chute **24** has a third reconveyance surface **24B** facing downward. The third reconveyance surface **24B** is a surface opposite to the sheet conveyance surface **24A** in the upper-lower direction, and faces the reconveyance path **P2**. The fuser chute **24** forms an upper surface of the reconveyance path **P2** by the third reconveyance surface **24B**. The third reconveyance surface **24B** guides the upper side of the sheet **S** conveyed along the reconveyance path **P2**.

[0089] In this way, the upper surface of the reconveyance path **P2** is formed by the third reconveyance surface **24B** of the fuser chute **24**. Thus, it is not necessary to provide a member forming the upper surface of the reconveyance path **P2** separately from the fuser chute **24**, which reduces the number of components. Further, the size of the apparatus main body **2** in the upper-lower direction is reduced.

[0090] As shown in FIGS. **6** to **8**, the image forming apparatus **1** includes a movable chute **71**. The movable chute **71** is located between the transfer unit **5A** and the fuser **6** in the front-rear direction. That is, the movable chute **71** is located downstream of the transfer unit **5A** and upstream of the fuser **6** in the first direction **D1**. The fuser chute **24** is located downstream of the movable chute **71** in the first direction **D1**.

[0091] The movable chute **71** is located on the opposite side of the reconveyance chute **23** with the reconveyance path **P2** interposed therebetween in the upper-lower direction, which is the thickness direction of the sheet **S** conveyed along the conveyance path **P1**. The movable chute **71** is located above the reconveyance chute **23**. The movable chute **71** is located below the conveyance path **P1**. That is, the movable chute **71** is located between the conveyance path **P1** and the reconveyance path **P2** in the upper-lower direction.

[0092] The movable chute **71** has a sheet conveyance surface **71A** facing upward and a first reconveyance surface **71B** facing downward. That is, the first reconveyance surface **71B** faces toward the opposite side of the sheet conveyance surface **71A** in the upper-lower direction. The

sheet conveyance surface **71A** guides, from below, the sheet **S** being conveyed along the conveyance path **P1** toward the fuser **6**. The first reconveyance surface **71B** guides, from above, the sheet **S** being conveyed along the reconveyance path **P2** toward the transfer unit **5A**. The first reconveyance surface **71B** forms an upper surface of the reconveyance path **P2**.

[0093] In this way, the upper surface of the reconveyance path **P2** is formed by the first reconveyance surface **71B** of the movable chute **71** located above the reconveyance chute **23**. Thus, it is not necessary to provide a member forming the upper surface of the reconveyance path **P2** separately from the movable chute **71**, which reduces the number of components. Further, the size of the apparatus main body **2** in the upper-lower direction is reduced.

[0094] In the image forming apparatus **1**, the movable chute **71** is located on the opposite side of the reconveyance chute **23** with the reconveyance path **P2** interposed therebetween, and the movable chute **71** forms the reconveyance path **P2**. Thus, when the movable chute **71** is moved to a second position described later, the movement range of the movable chute **71** is not restricted. This secures a large space in which the conveyance path **P1** is located (that is, a large space for the conveyance path **P1**), and the sheet **S** is conveyed efficiently. Further, the number of components constituting the apparatus main body **2** is reduced, and the apparatus main body **2** is downsized in the upper-lower direction.

[0095] In particular, the first reconveyance surface **71B** forms the reconveyance path **P2** when the movable chute **71** is located at the second position. Thus, the reconveyance path **P2** is formed by the movable chute **71** in a state where a large space for the conveyance path **P1** is secured.

[0096] The movable chute **71** is pivotable about a pivot shaft **713** located at the front end. The movable chute **71** is pivotable about the pivot shaft **713** to be movable between a first position (position shown in FIG. 7) and the second position (position shown in FIG. 8) at which a downstream end **71a** of the movable chute **71** in the first direction **D1** is farther from the conveyance path **P1** compared to when the movable chute **71** is located at the first position. When the movable chute **71** moves from the first position to the second position, the downstream end **71a** of the movable chute **71** in the first direction **D1** contacts the sheet conveyance surface **24A** of the fuser chute **24**, thereby positioning the movable chute **71** at the second position.

[0097] When the movable chute **71** is located at the first position, the sheet conveyance surface **71A** of the movable chute **71** guides the sheet **S** conveyed from the transfer unit **5A** toward the fuser **6**. When the movable chute **71** is located at the second position, the sheet conveyance surface **71A** is located farther away from the conveyance path **P1** compared to when the movable chute **71** is located at the first position.

[0098] When the sheet **S** is conveyed by the photosensitive drum **54** and the transfer roller **55** of the transfer unit **5A** and by the fuser **6**, the conveyance speed of the sheet **S** by the transfer unit **5A** is higher than the conveyance speed of the sheet **S** by the fuser **6**, and the sheet **S** between the fuser **6** and the transfer unit **5A** is bent (curved). When the movable chute **71** is located at the second position, the sheet conveyance surface **71A** guides the sheet **S** bent (curved) toward the sheet conveyance surface **71A** between the fuser **6** and the transfer unit **5A**.

[0099] In this way, the sheet **S** is guided by the movable chute **71** located at the first position closer to the conveyance path **P1** than at the second position. Thus, the sheet conveyed from the transfer unit **5A** constituted by the photosensitive drum **54** and the transfer roller **55** is appropriately guided to the fuser **6** by the movable chute **71**.

[0100] The position of the downstream end **71a** of the movable chute **71** when the movable chute **71** is located at the second position is lower than the position of the downstream end **71a** of the movable chute **71** when the movable chute **71** is located at the first position. Thus, when the movable chute **71** moves to the second position, a large space is secured in which the conveyance path **P1** between the transfer unit **5A** and the fuser **6** is located, which prevents the bent (curved) sheet **S** from interfering with (contacting) other components of the apparatus main body **2**.

[0101] The first reconveyance surface **71B** of the movable chute **71** is located closer to the

reconveyance path P2 when the movable chute **71** is located at the second position than when the movable chute **71** is located at the first position. When the movable chute **71** is located at the second position, the first reconveyance surface **71B** forms an upper surface of the reconveyance path P2. When the movable chute **71** is located at the second position, the first reconveyance surface **71B** guides the upper side of the sheet S conveyed along the reconveyance path P2.

[0102] As shown in FIGS. **9** to **12**, the movable chute **71** includes a base portion **711**, a plurality of guide ribs **712**, and the pivot shaft **713**. The movable chute **71** is made of a material containing, for example, polyethylene terephthalate (PET) and glass resin.

[0103] The base portion **711** is formed of a plate-like member curved so as to be recessed downward when viewed in the left-right direction, and extends in the front-rear direction and the left-right direction. Each guide rib **712** protrudes upward from the upper surface of the base portion **711** and extends in the front-rear direction. The plurality of guide ribs **712** are arranged at intervals along the left-right direction. The front ends of the guide ribs **712** protrude forward from the base portion **711**. The rear ends of the guide ribs **712** protrude rearward from the base portion **711**. The upper surfaces of the guide ribs **712** form the sheet conveyance surface **71A** of the movable chute **71**.

[0104] The lower surface of the base portion **711** forms the first reconveyance surface **71B** of the movable chute **71**. The first reconveyance surface **71B** has a rear end **71Ba** which is an upstream end in the second direction D2 and a front end **71Bb** which is a downstream end in the second direction D2.

[0105] The pivot shafts **713** are located at the front end of the base portion **711**. The pivot shafts **713** are provided at left and right ends of the base portion **711**.

[0106] A chute sheet metal **72** is placed on the upper surface of the movable chute **71**. The chute sheet metal **72** is supported by the movable chute **71**. The chute sheet metal **72** is formed of a metal member and is connected to ground. The chute sheet metal **72** is an example of a sheet metal member. The chute sheet metal **72** includes a base portion **721**, an upstream flange **722**, a downstream flange **723**, and a plurality of slits **724**.

[0107] The base portion **721** is formed of a plate-like member curved along the curved base portion **711** of the movable chute **71**. The upstream flange **722** extends downward from a front end of the base portion **721**, which is an upstream end of the base portion **721** in the first direction D1, and then extends forward. The downstream flange **723** extends downward from a rear end of the base portion **721**, which is a downstream end of the base portion **721** in the first direction D1.

[0108] The plurality of slits **724** are formed, and each of the slits extends in the front-rear direction. The plurality of slits **724** are arranged at intervals along the left-right direction. Each slit **724** is a hole through which the guide rib **712** of the movable chute **71** passes from below. The chute sheet metal **72** is superposed on the upper surface of the movable chute **71** in a state where the guide ribs **712** of the movable chute **71** pass through the slits **724** from below. By superposing the chute sheet metal **72** on the movable chute **71**, the rigidity of the movable chute **71** is increased by the chute sheet metal **72**.

[0109] The grounded chute sheet metal **72** is superposed on the movable chute **71**. Thus, when the sheet S charged by passing through the transfer unit 5A is guided by the movable chute **71**, the sheet S is attracted to the chute sheet metal **72**, which prevents the sheet S from moving around (flapping) while being conveyed. Thus, the sheet S is stably conveyed.

[0110] The front end of the slit **724** is engaged with the front end of the guide rib **712**, and the rear end of the slit **724** is engaged with the rear end of the guide rib **712**. This prevents the chute sheet metal **72** superposed on the movable chute **71** from moving upward with respect to the movable chute **71**.

[0111] The movable chute **71** and the chute sheet metal **72** are fixed by a chute stopper **73**. The chute stopper **73** is an example of a stopper. The chute stopper **73** is formed of a plate-like member. The chute stopper **73** is formed of, for example, an ABS resin. The chute stopper **73** is attached to a

position at a rear end and a left-right center of the movable chute **71**. The chute stopper **73** is attached to the movable chute **71** from the first reconveyance surface **71B** side of the movable chute **71**.

[0112] As shown in FIGS. **11A** and **11B**, the first reconveyance surface **71B** of the movable chute **71** includes a central reconveyance surface **71BC**, a left reconveyance surface **71BL**, and a right reconveyance surface **71BR**. The central reconveyance surface **71BC** is located at the center of the first reconveyance surface **71B** in the left-right direction. The left reconveyance surface **71BL** is located to the left of the central reconveyance surface **71BC**. The right reconveyance surface **71BR** is located to the right of the central reconveyance surface **71BC**.

[0113] The rear end **71Ba** of the central reconveyance surface **71BC** is recessed forward from the rear ends **71Ba** of the left reconveyance surface **71BL** and the right reconveyance surface **71BR**. A stopper attachment surface **71C** is formed on the base portion **711** at a position rearward of the rear end **71Ba** of the central reconveyance surface **71BC** (see FIG. **12**). The stopper attachment surface **71C** faces downward. The stopper attachment surface **71C** is located above the central reconveyance surface **71BC**.

[0114] As shown in FIGS. **11A**, **11B** and **12**, the movable chute **71** includes a holding portion **715**. The holding portion **715** is disposed on the stopper attachment surface **71C** of the base portion **711**. The holding portion **715** includes a first wall **715a** that protrudes downward from the stopper attachment surface **71C** of the base portion **711**, and a second wall **715b** that protrudes rearward from the lower end of the first wall **715a**. The holding portions **715** are provided at left and right positions (two positions) of the stopper attachment surface **71C**, and hold the left end and the right end of the chute stopper **73**.

[0115] The chute stopper **73** attached to the movable chute **71** is sandwiched between the base portion **711** and the second wall **715b** of the holding portion **715** in the upper-lower direction. In this way, the chute stopper **73** is held by the holding portion **715**. The chute stopper **73** attached to the movable chute **71** covers the stopper attachment surface **71C** from below.

[0116] The chute stopper **73** attached to the movable chute **71** is sandwiched between the downstream flange **723** of the chute sheet metal **72** and the first wall **715a** of the holding portion **715** in the front-rear direction. This prevents movement of the chute sheet metal **72** in the front-rear direction with respect to the movable chute **71**, and the chute sheet metal **72** is fixed with respect to the movable chute **71**.

[0117] A chute spring **74** engages with the left and right pivot shafts **713** of the movable chute **71**. The chute spring **74** is a torsion spring. The chute spring **74** urges the movable chute **71** in a direction from the second position toward the first position.

[0118] As shown in FIGS. **6** to **8**, the image forming apparatus **1** includes a guide frame **25**. The guide frame **25** is located upstream of the movable chute **71** in the first direction **D1**, and pivotably supports the movable chute **71**. The guide frame **25** is located below the transfer unit **5A**. The guide frame **25** rotatably supports the registration roller **35a**.

[0119] The guide frame **25** is located above the reconveyance path **P2** and forms an upper surface of the reconveyance path **P2**. The guide frame **25** has a second reconveyance surface **25A** facing downward. The second reconveyance surface **25A** faces the reconveyance path **P2**. The guide frame **25** forms an upper surface of the reconveyance path **P2** by the second reconveyance surface **25A**. The second reconveyance surface **25A** guides the upper side of the sheet **S** conveyed along the reconveyance path **P2**.

[0120] In this way, the upper surface of the reconveyance path **P2** is formed by the second reconveyance surface **25A** of the guide frame **25**. Thus, it is not necessary to provide a member for forming the upper surface of the reconveyance path **P2** separately from the guide frame **25**, which reduces the number of components. Further, the size of the apparatus main body **2** in the upper-lower direction is reduced.

[0121] Positional relationships among the fuser chute, the movable chute, and the guide frame will

be described. As shown in FIGS. 13 to 16, the third reconveyance surface 24B of the fuser chute 24 includes a central reconveyance surface 24BC, a left reconveyance surface 24BL, and a right reconveyance surface 24BR. The central reconveyance surface 24BC is located at the center of the third reconveyance surface 24B in the left-right direction. The left reconveyance surface 24BL is located to the left of the central reconveyance surface 24BC. The right reconveyance surface 24BR is located to the right of the central reconveyance surface 24BC.

[0122] Each of the central reconveyance surface 24BC, the left reconveyance surface 24BL, and the right reconveyance surface 24BR has a front end 24Ba which is a downstream end of the third reconveyance surface 24B of the fuser chute 24. The front end 24Ba of the central reconveyance surface 24BC protrudes forward from the front ends 24Ba of the left reconveyance surface 24BL and the right reconveyance surface 24BR.

[0123] The front ends 24Ba of the left reconveyance surface 24BL and the right reconveyance surface 24BR of the third reconveyance surface 24B of the fuser chute 24 are located rearward of the rear end 71Ba of the first reconveyance surface 71B of the movable chute 71. In contrast, the front end 24Ba of the central reconveyance surface 24BC of the third reconveyance surface 24B of the fuser chute 24 is located forward of the rear ends 71Ba of the left reconveyance surface 71BL and the right reconveyance surface 71BR of the movable chute 71.

[0124] That is, a portion of the fuser chute 24 where the central reconveyance surface 24BC is formed and portions of the movable chute 71 where the left reconveyance surface 71BL and the right reconveyance surface 71BR are formed overlap each other in the second direction D2. Thus, when the sheet S conveyed along the reconveyance path P2 is guided by the third reconveyance surface 24B of the fuser chute 24 and the first reconveyance surface 71B of the movable chute 71, the sheet S is guided seamlessly by the fuser chute 24 and the movable chute 71. Thus, the sheet S conveyed along the reconveyance path P2 is smoothly handed over from the fuser chute 24 to the movable chute 71.

[0125] The front end 24Ba of the central reconveyance surface 24BC of the fuser chute 24 is located rearward of the rear end 71Ba of the central reconveyance surface 71BC of the movable chute 71.

[0126] A front end portion of the central reconveyance surface 24BC of the third reconveyance surface 24B of the fuser chute 24 is located below the stopper attachment surface 71C of the movable chute 71, and covers the chute stopper 73 attached to the stopper attachment surface 71C from below. That is, the movable chute 71 and the chute stopper 73 overlap each other when viewed in the upper-lower direction.

[0127] Thus, even when the chute stopper 73 is attached to the movable chute 71, the sheet S conveyed along the reconveyance path P2 does not interfere with (contact) the chute stopper 73 when the sheet S is conveyed from the third reconveyance surface 24B to the first reconveyance surface 71B. Thus, the sheet S is smoothly handed over from the fuser chute 24 to the movable chute 71.

[0128] The central reconveyance surface 24BC of the third reconveyance surface 24B of the fuser chute 24, and the left reconveyance surface 71BL and the right reconveyance surface 71BR of the first reconveyance surface 71B of the movable chute 71 overlap each other when viewed in the left-right direction.

[0129] Thus, when the sheet S conveyed along the reconveyance path P2 is guided by the third reconveyance surface 24B and the first reconveyance surface 71B, the sheet S is guided by both the third reconveyance surface 24B and the first reconveyance surface 71B at a boundary between the third reconveyance surface 24B and the first reconveyance surface 71B. Thus, the sheet S conveyed along the reconveyance path P2 is smoothly handed over from the fuser chute 24 to the movable chute 71.

[0130] As shown in FIGS. 15 and 16, in a state where the movable chute 71 is located at the second position, a rear end 71Ba of the movable chute 71, which is an upstream end of the first

reconveyance surface **71B** in the second direction **D2**, is located farther from the reconveyance path **P2** than the front end **24Ba** of the fuser chute **24**, which is a downstream end of the third reconveyance surface **24B** in the second direction **D2**. That is, the rear end **71Ba** of the first reconveyance surface **71B** is located above the front end **24Ba** of the third reconveyance surface **24B**.

[0131] Accordingly, a step is formed at the boundary between the third reconveyance surface **24B** and the first reconveyance surface **71B**, the step being that the first reconveyance surface **71B** is farther away from the reconveyance path **P2** than the third reconveyance surface **24B**. Thus, the sheet **S** conveyed along the reconveyance path **P2** is not caught at the boundary between the third reconveyance surface **24B** of the fuser chute **24** and the first reconveyance surface **71B** of the movable chute **71**, and the sheet **S** is smoothly conveyed.

[0132] In a state where the movable chute **71** is located at the second position, a rear end **25Aa** of the guide frame **25**, which is the upstream end of the second reconveyance surface **25A** in the second direction **D2**, is located farther away from the reconveyance path **P2** than the front end **71Bb** of the movable chute **71**, which is the downstream end of the first reconveyance surface **71B** in the second direction **D2**. That is, the rear end **25Aa** of the second reconveyance surface **25A** is located above the front end **71Bb** of the first reconveyance surface **71B**.

[0133] Accordingly, a step is formed at the boundary between the first reconveyance surface **71B** and the second reconveyance surface **25A**, the step being that the second reconveyance surface **25A** is farther away from the reconveyance path **P2** than the first reconveyance surface **71B**. Thus, the sheet **S** conveyed along the reconveyance path **P2** is not caught at the boundary between the first reconveyance surface **71B** of the movable chute **71** and the second reconveyance surface **25A** of the guide frame **25**, and the sheet **S** is smoothly conveyed.

[0134] Movement control of the movable chute will be described. As shown in FIG. 17, the image forming apparatus **1** includes a solenoid **92** as a drive source for moving the movable chute **71**. The solenoid **92** is a solenoid actuator for moving the movable chute **71**, which is urged toward the first position by the chute spring **74**, from the first position toward the second position. The solenoid **92** is connected to the controller **95**, and the controller **95** controls the operation of the solenoid **92**. The controller **95** controls the movement of the movable chute **71** by controlling the operation of the solenoid **92**. The pre-registration sensor **97**, the post-registration sensor **98**, and the sheet sensor **99** are connected to the controller **95**.

[0135] After the leading edge of the sheet **S** conveyed along the conveyance path **P1** passes the fuser **6**, the controller **95** controls the solenoid **92** to move the movable chute **71** from the first position to the second position. The solenoid **92** is configured to move the movable chute **71** via a chute link **75**, for example.

[0136] In this case, the controller **95** may determine that the leading edge of the sheet **S** has passed the fuser **6** when a particular time has elapsed since the post-registration sensor **98** detects the leading edge of the sheet **S**, and control the solenoid **92** to move the movable chute **71** from the first position to the second position. For example, when the solenoid **92** is driven by the controller **95** and is turned on, the movable chute **71** is pressed by the chute link **75** and moves from the first position to the second position against the urging force of the chute spring **74**.

[0137] After the leading edge of the sheet **S** passes the fuser **6**, the sheet **S** between the transfer unit **5A** and the fuser **6** is bent (curved) because the conveyance speed of the sheet **S** by the transfer unit **5A** is higher than the conveyance speed of the sheet **S** by the fuser **6**. Thus, by moving the movable chute **71** toward the second position by the control of the controller **95**, a large space is secured for a space in which the conveyance path **P1** between the transfer unit **5A** and the fuser **6** is located. This prevents the sheet **S** in which the bending occurs from interfering with (contacting) other components of the apparatus main body **2**.

[0138] After the movable chute **71** is moved from the first position to the second position by the control of the controller **95**, the movable chute **71** is maintained at the second position while the

leading edge of the sheet S having passed the fuser 6 passes through the reconveyance path P2. Further, after the leading edge of the sheet S conveyed along the reconveyance path P2 passes the registration roller 35a, the solenoid 92 is controlled to move the movable chute 71 from the second position toward the first position.

[0139] In this case, for example, when a particular time has elapsed after the pre-registration sensor 97 detects the leading edge of the sheet S, the controller 95 may determine that the leading edge of the sheet S has passed the registration roller 35a and control the solenoid 92 to move the movable chute 71 from the second position toward the first position. For example, when the solenoid 92 becomes an OFF state by stopping the driving of the solenoid 92 by the controller 95, the pressing of the solenoid 92 is released, and the movable chute 71 moves from the second position to the first position by the urging force of the chute spring 74.

[0140] In a case where the movable chute 71 is moved from the first position toward the second position after the leading edge of the sheet S passes the fuser 6, after the state where the sheet S is conveyed by both the fuser 6 and the transfer unit 5A is released, the bending of the sheet S is eliminated. Thus, the movable chute 71 may be at the first position during a period from when the state where the sheet S is conveyed by both the fuser 6 and the transfer unit 5A is released until when the leading edge of the sheet S reaches the first reconveyance surface 71B of the movable chute 71.

[0141] However, when the movable chute 71 is moved to the second position, the reconveyance path P2 is formed by the first reconveyance surface 71B of the movable chute 71. Thus, the movable chute 71 may be maintained at the second position during a period from when the movable chute 71 is moved to the second position until the leading edge of the sheet S passes the registration roller 35a after the sheet S is conveyed along the reconveyance path P2. In this way, by maintaining the movable chute 71 at the second position until the leading edge of the sheet S passes the registration roller 35a, the movement control of the movable chute 71 is simplified.

[0142] A modification of movement control of the movable chute will be described. The controller 95 may control the movement of the movable chute 71 in the following manner.

[0143] After the leading edge of the sheet S conveyed along the conveyance path P1 passes the fuser 6, the controller 95 controls the solenoid 92 to move the movable chute 71 from the first position to the second position.

[0144] In this case, the controller 95 may determine that the leading edge of the sheet S has passed the fuser 6 when a particular time has elapsed since the post-registration sensor 98 detects the leading edge of the sheet S, and control the solenoid 92 to move the movable chute 71 from the first position to the second position.

[0145] Next, after the trailing edge of the sheet S conveyed along the conveyance path P1 passes the transfer unit 5A, the controller 95 controls the solenoid 92 to move the movable chute 71 from the second position to the first position.

[0146] In this case, the controller 95 may determine that the trailing edge of the sheet S has passed the transfer unit 5A when a particular time has elapsed since the post-registration sensor 98 detects the trailing edge of the sheet S, and control the solenoid 92 to move the movable chute 71 from the second position to the first position.

[0147] Further, before the leading edge of the sheet S conveyed along the reconveyance path P2 after passing through the fuser 6 and reversing the conveyance direction reaches the first reconveyance surface 71B of the movable chute 71, the controller 95 controls the solenoid 92 to move the movable chute 71 from the first position to the second position.

[0148] In this case, when a particular time has elapsed since the sheet sensor 99 detects the leading edge (before reversing the conveyance direction) of the sheet S, the controller 95 determines that the leading edge (after reversing the conveyance direction) of the sheet S reaches the first reconveyance surface 71B. Thus, before the particular time elapses since the sheet sensor 99 detects the leading edge of the sheet S, the controller 95 may control the solenoid 92 to move the

movable chute **71** from the first position to the second position.

[0149] Then, after the leading edge of the sheet **S** conveyed along the reconveyance path **P2** passes the registration roller **35a**, the controller **95** controls the solenoid **92** to move the movable chute **71** from the second position to the first position.

[0150] In this case, for example, the controller **95** may determine that the leading edge of the sheet **S** has passed the registration roller **35a** when a particular time has elapsed after the pre-registration sensor **97** detects the leading edge of the sheet **S**, and control the solenoid **92** to move the movable chute **71** from the second position to the first position.

[0151] In a configuration in which the movable chute **71** is moved to the second position by the solenoid **92**, the solenoid **92** is in an ON state when the movable chute **71** is at the second position. Thus, if the movable chute **71** is continuously maintained at the second position during a period from when the leading edge of the sheet **S** passes the fuser **6** to when the leading edge of the sheet **S** passes through the reconveyance path **P2**, heat is generated from the solenoid **92** and the amount of current consumed by the solenoid **92** increases.

[0152] Thus, when the movable chute **71** does not need to be at the second position after the leading edge of the sheet **S** passes the fuser **6**, the movable chute **71** is moved to the first position, which reduces the heat generation from the solenoid **92** and the current consumption by the solenoid **92**.

[0153] As shown in FIGS. **17** to **19**, the chute link **75** is connected to the solenoid **92**. The chute link **75** is driven by the solenoid **92** to press the movable chute **71**, thereby moving the movable chute **71** from the first position to the second position.

[0154] The solenoid **92** includes a main body **921** supported by the second main body frame **27** and a plunger **922** movable in the left-right direction with respect to the main body **921**. The plunger **922** is urged rightward from the main body **921** by an urging member. In an OFF state in which the solenoid **92** is not driven by the controller **95**, the plunger **922** is in a state of being moved rightward and protruding from the main body **921**. In an ON state in which the solenoid **92** is driven by the controller **95**, the plunger **922** is pulled into the main body **921** and moved leftward.

[0155] The chute link **75** includes a first link **751**, a second link **752**, and a third link **753**. The first link **751** and the second link **752** are located inside the air duct **28**. The third link **753** protrudes downward from the inside of the air duct **28**. A holder **77** for supporting the chute link **75** is attached to the inside of the air duct **28**.

[0156] The first link **751** is an elongated member extending in the left-right direction and is supported by the holder **77** so as to be movable in the left-right direction. The first link **751** has a first end portion **751a** which is a left end portion and a second end portion **751b** which is a right end portion. The first end portion **751a** is connected to the plunger **922** of the solenoid **92**. The second end portion **751b** contacts the second link **752**.

[0157] The second link **752** is an elongated member extending in the left-right direction and is located below the first link **751**. The second link **752** includes a pivot shaft portion **752a** located at a right end, a contact portion **752b** extending upward from the pivot shaft portion **752a**, a tip end portion **752c** which is a left end, and an engagement portion **752d** extending obliquely upward and rightward from the pivot shaft portion **752a**. The second link **752** is supported by the holder **77** so as to be pivotable about the pivot shaft portion **752a**.

[0158] The image forming apparatus **1** includes a link spring **76** that urges the second link **752**. The link spring **76** is attached to the holder **77**. One end of the link spring **76** is engaged with a hook **771** of the holder **77**, and the other end of the link spring **76** is engaged with the engagement portion **752d** of the second link **752**. The second link **752** is urged by the link spring **76** so as to pivot in a direction in which the tip end portion **752c** moves upward. The contact portion **752b** contacts the second end portion **751b** of the first link **751**. When the first link **751** moves leftward, the contact portion **752b** is pressed leftward by the second end portion **751b**, and the second link **752** pivots in a direction in which the tip end portion **752c** moves downward.

[0159] The third link **753** is an elongated member extending in the upper-lower direction, and is disposed so as to be movable in the upper-lower direction. The third link **753** has a connecting portion **753a** located at the upper end and a pressing portion **753b** located at the lower end. The connecting portion **753a** is connected to the tip end portion **752c** of the second link **752**. The pressing portion **753b** is configured to press a pressed portion **714** of the movable chute **71**.

[0160] As shown in FIGS. **9**, **18**, and **19**, the pressed portion **714** of the movable chute **71** is formed on the sheet conveyance surface **71A** side and faces upward. The sheet conveyance surface **71A** of the movable chute **71** is formed in a region where the sheet **S** passes with respect to the left-right direction, and the pressed portion **714** is located outside the region where the sheet conveyance surface **71A** is formed with respect to the left-right direction. In the present embodiment, the pressed portion **714** is located on the left side of the region where the sheet conveyance surface **71A** is formed.

[0161] The pressing portion **753b** of the third link **753** and the pressed portion **714** of the movable chute **71** face each other in the upper-lower direction. When the third link **753** moves downward, the pressing portion **753b** presses the pressed portion **714** from above.

[0162] In the chute link **75** configured as described above, in an OFF state in which the solenoid **92** is not driven by the controller **95**, the first link **751** connected to the plunger **922** is moved rightward, and the second link **752** is rotated in a direction in which the tip end portion **752c** is moved upward by the urging force of the link spring **76**.

[0163] The third link **753** is in a state of being moved upward by the second link **752**, and the pressing portion **753b** of the third link **753** and the pressed portion **714** of the movable chute **71** are separated from each other. The movable chute **71** is at the first position by the urging force of the chute spring **74**.

[0164] In an ON state in which the solenoid **92** is driven by the controller **95**, the first link **751** is moved leftward by the plunger **922** pulled by the main body **921**. When the first link **751** moves leftward, the contact portion **752b** of the second link **752** is pressed by the second end portion **751b** of the first link **751**, and the second link **752** pivots in a direction in which the tip end portion **752c** moves downward against the urging force of the link spring **76**.

[0165] The third link **753** moves downward in accordance with the pivotal movement of the second link **752**, and the pressing portion **753b** of the third link **753** presses the pressed portion **714** of the movable chute **71** downward, whereby the movable chute **71** moves from the first position to the second position against the urging force of the chute spring **74**.

[0166] As described above, the chute link **75** is configured to move the movable chute **71** from the first position to the second position by being driven by the solenoid **92** in the ON state and pressing the movable chute **71** from above, that is, from the sheet conveyance surface **71A** side. Thus, the movable chute **71** is moved from the first position to the second position quickly and stably.

[0167] Conversely, when the solenoid **92** in the ON state is switched to the OFF state, the chute link **75** is separated from the movable chute **71**, and the movable chute **71** is moved from the second position to the first position by the urging force of the chute spring **74**.

[0168] A second embodiment of the movable chute, the chute sheet metal, and the fuser chute will be described. The movable chute **71**, the chute sheet metal **72**, and the fuser chute **24** may be configured as a movable chute **171**, a chute sheet metal **172**, and a fuser chute **124** shown in FIGS. **20A**, **20B**, **21** and **22**.

[0169] The movable chute **171** and the chute sheet metal **172** are different from the movable chute **71** and the chute sheet metal **72** which are fixed to each other by using the chute stopper **73**, in that the movable chute **171** and the chute sheet metal **172** are fixed to each other without using a chute stopper.

[0170] The movable chute **171** includes a base **1711**, guide ribs **1712**, pivot shafts **1713**, a pressed portion **1714**, and hooks **1715**. The hooks **1715** are formed on the upper surface of the base **1711**.

[0171] The base **1711** is different from the base portion **711** in that the hooks **1715** are formed on

the upper surface and the chute stopper **73** is not attached, and the other configurations are the same as those of the base portion **711**.

[0172] The guide ribs **1712**, the pivot shafts **1713**, and the pressed portion **1714** are formed in the same manner as the guide ribs **712**, the pivot shafts **713**, and the pressed portion **714** of the movable chute **71**, respectively. The upper surfaces of the guide ribs **1712** form a sheet conveyance surface **171A** of the movable chute **171**. The lower surface of the base **1711** forms a first reconveyance surface **171B** of the movable chute **171**.

[0173] The chute sheet metal **172** includes a base portion **1721**, an upstream flange **1722**, a downstream flange **1723**, slits **1724**, and engagement holes **1725**. The base portion **1721**, the upstream flange **1722**, the downstream flange **1723**, and the slits **1724** are formed in the same manner as the base portion **721**, the upstream flange **722**, the downstream flange **723**, and the slits **724** of the chute sheet metal **72**, respectively. The engagement holes **1725** are formed in the base portion **1721** and configured to engage with the hooks **1715** of the movable chute **171**.

[0174] The chute sheet metal **172** is fixed to the movable chute **171** by engaging the hooks **1715** with the engaging holes **1725** in a state where the chute sheet metal **172** is superposed on the upper surface of the movable chute **171**.

[0175] The first reconveyance surface **171B** has a rear end **171Ba** which is an upstream end in the second direction **D2** and a front end **171Bb** which is a downstream end in the second direction **D2**. Since no chute stopper is disposed on the first reconveyance surface **171B** of the movable chute **171**, the rear end **171Ba** of the first reconveyance surface **171B** is located at the same position in the front-rear direction over the entirety in the left-right direction.

[0176] The fuser chute **124** has a sheet conveyance surface **124A** and a third reconveyance surface **124B**. The fuser chute **124** is different from the fuser chute **24** in that a front end **124Ba** of the third reconveyance surface **124B** is located at the same position in the front-rear direction over the entirety in the left-right direction. In the fuser chute **24**, the front end **24Ba** of the central reconveyance surface **24BC** protrudes forward from the front ends **24Ba** of the left reconveyance surface **24BL** and the right reconveyance surface **24BR**. The other configuration of the fuser chute **124** is the same as that of the fuser chute **24**.

[0177] The front end **124Ba** of the third reconveyance surface **124B** of the fuser chute **124** is located rearward of the rear end **171Ba** of the first reconveyance surface **171B**, and the fuser chute **124** and the movable chute **171** do not overlap each other in the front-rear direction.

[0178] No chute stopper is attached to the first reconveyance surface **171B** of the movable chute **171**, and the fuser chute **124** does not need to cover the lower side of the first reconveyance surface **171B**. Thus, the sheet **S** is smoothly conveyed even though the fuser chute **124** and the movable chute **171** do not overlap in the front-rear direction.

[0179] The movable chute **171** is formed such that the rear end **171Ba** of the first reconveyance surface **171B** is located at the same position in the front-rear direction over the entirety in the left-right direction, and thus the shape of the movable chute **171** is simplified. The fuser chute **124** is formed such that the front end **124Ba** of the third reconveyance surface **124B** is located at the same position in the front-rear direction over the entirety in the left-right direction, and thus the shape of the fuser chute **124** is simplified.

[0180] In the movable chute **171** and the fuser chute **124**, the rear end **171Ba** of the first reconveyance surface **171B** of the movable chute **171** is located farther from the reconveyance path **P2** compared to the front end **124Ba** of the third reconveyance surface **124B** of the fuser chute **124**. Thus, the sheet **S** is smoothly conveyed between the fuser chute **124** and the movable chute **171**.

[0181] In the guide frame **25** and the movable chute **171**, the rear end **25Aa** of the second reconveyance surface **25A** of the guide frame **25** is located farther from the reconveyance path **P2** compared to the front end **171Bb** of the first reconveyance surface **171B** of the movable chute **171**. Thus, the sheet **S** is smoothly conveyed between the movable chute **171** and the guide frame **25**.

[0182] A third embodiment of the movable chute will be described. The movable chute **71** may be

formed as a movable chute **271** shown in FIGS. **23** and **24**.

[0183] The movable chute **271** has a sheet conveyance surface **271A** facing upward and a first reconveyance surface **271B** facing downward on the opposite side of the sheet conveyance surface **271A** in the upper-lower direction. The sheet conveyance surface **271A** guides, from below, the sheet **S** being conveyed along the conveyance path **P1** toward the fuser **6**. The first reconveyance surface **271B** guides, from above, the sheet **S** being conveyed along the reconveyance path **P2** toward the transfer unit **5A**. The first reconveyance surface **271B** forms an upper surface of the reconveyance path **P2**.

[0184] The movable chute **271** is pivotable about a pivot shaft **2713** located at the front end. The movable chute **271** is pivotable about the pivot shaft **2713** to be movable between a first position (position shown in FIG. **23**) and a second position (position shown in FIG. **24**) at which a downstream end **271a** of the movable chute **271** in the first direction **D1** is farther from the conveyance path **P1** than at the first position.

[0185] The sheet conveyance surface **271A** of the movable chute **271** guides the sheet **S** conveyed from the transfer unit **5A** toward the fuser **6** when the movable chute **271** is located at the first position. The first reconveyance surface **271B** of the movable chute **271** forms the reconveyance path **P2** when the movable chute **271** is located at the first position.

[0186] When the movable chute **271** is located at the second position, the sheet conveyance surface **271A** is located farther from the conveyance path **P1** than when the movable chute **271** is located at the first position. When the movable chute **271** is located at the second position, the sheet conveyance surface **271A** guides the sheet **S** bent (curved) toward the sheet conveyance surface **271A** between the fuser **6** and the transfer unit **5A**.

[0187] The position of the downstream end **271a** of the movable chute **271** when the movable chute **271** is located at the second position is lower than the position of the downstream end **271a** of the movable chute **271** when the movable chute **271** is located at the first position. Thus, when the movable chute **271** moves to the second position, a large space is secured in which the conveyance path **P1** between the transfer unit **5A** and the fuser **6** is located, which prevents the bent (curved) sheet **S** from interfering with (contacting) other components of the apparatus main body **2**.

[0188] When the movable chute **271** is located at the second position, the first reconveyance surface **271B** enters the reconveyance path **P2**. By configuring the first reconveyance surface **271B** to enter the reconveyance path **P2** when the movable chute **271** is located at the second position, the sheet conveyance surface **271A** is moved to a position separated from the conveyance path **P1** when the movable chute **271** is moved to the second position. This secures a space where the conveyance path **P1** is located as large as possible in a limited space, and effectively absorbs bending (deflection) of the sheet **S**.

[0189] The movable chute **271** is configured such that the first reconveyance surface **271B** forms the reconveyance path **P2** when the movable chute **271** is located at the first position.

[0190] In this configuration, after the leading edge of the sheet **S** conveyed from the transfer unit **5A** along the conveyance path **P1** passes the fuser **6**, the movable chute **271** is moved from the first position to the second position. Then, after the trailing edge of the sheet **S** passes the transfer unit **5A** and before the leading edge of the sheet **S** reaches the first reconveyance surface **271B**, the movable chute **271** is moved from the second position to the first position. By moving the movable chute **271** in this way, the sheet **S** is conveyed smoothly and efficiently.

[0191] In this configuration, the movable chute **271** may be configured to be moved without using the solenoid **92** and the chute link **75**. In this case, the movable chute **271** may be moved from the first position to the second position by using force of the bent (curved) sheet **S** that presses the movable chute **271**, the sheet **S** being conveyed from the transfer unit **5A** along the conveyance path **P1**. Further, when the trailing edge of the sheet **S** passes the transfer unit **5A**, the bending of the sheet **S** is eliminated, and the pressing force of the sheet **S** against the movable chute **271** is eliminated, which causes the movable chute **271** to be moved from the second position to the first

position by the chute spring 74. Thus, the sheet S is smoothly and efficiently conveyed without performing the movement control of the movable chute 271.

[0192] While the present disclosure has been described in conjunction with various example structures outlined above and illustrated in the figures, various alternatives, modifications, variations, improvements, and/or substantial equivalents, whether known or that may be presently unforeseen, may become apparent to those having at least ordinary skill in the art. Accordingly, the example embodiments of the disclosure, as set forth above, are intended to be illustrative of the present disclosure, and not limiting the present disclosure. Various changes may be made without departing from the spirit and scope of the disclosure. Thus, the disclosure is intended to embrace all known or later developed alternatives, modifications, variations, improvements, and/or substantial equivalents. Some specific examples of potential alternatives, modifications, or variations in the described disclosure are provided as appropriate.

Claims

1. An image forming apparatus comprising: a transfer unit configured to transfer a toner image onto a sheet; a fuser including a heater and configured to fix the toner image on the sheet conveyed from the transfer unit; a movable chute located downstream of the transfer unit and upstream of the fuser in a first direction, the first direction being a sheet conveyance direction from the transfer unit toward the fuser, the movable chute having a downstream end in the first direction, the movable chute being movable between a first position and a second position different from the first position, the downstream end of the movable chute located at the second position being farther from a sheet conveyance path than the downstream end of the movable chute located at the first position, the sheet conveyance path being formed along the first direction; and a reconveyance chute configured to guide the sheet having passed the fuser to the transfer unit, the reconveyance chute forming a reconveyance path of the sheet that is reconveyed from the fuser toward the transfer unit, the reconveyance path being located between the movable chute and the reconveyance chute in a sheet thickness direction of the sheet, the movable chute having a sheet conveyance surface and a first reconveyance surface, the sheet conveyance surface being configured to guide the sheet that is conveyed from the transfer unit toward the fuser along the sheet conveyance path, the first reconveyance surface facing opposite to the sheet conveyance surface in the sheet thickness direction, the first reconveyance surface forming the reconveyance path.
2. The image forming apparatus according to claim 1, wherein the first reconveyance surface forms the reconveyance path when the movable chute is located at the second position.
3. The image forming apparatus according to claim 2, further comprising: a drive source configured to move the movable chute; and a controller configured to control the drive source, wherein the controller is configured to: after a leading edge of the sheet conveyed along the sheet conveyance path passes the fuser, control the drive source to move the movable chute from the first position toward the second position.
4. The image forming apparatus according to claim 3, further comprising: a registration roller located upstream of the transfer unit in the first direction, the registration roller being configured to correct a skew of the sheet that is conveyed toward the transfer unit, wherein the controller is configured to: after the leading edge of the sheet conveyed along the sheet conveyance path passes the fuser, control the drive source to move the movable chute from the first position toward the second position; while a leading edge of the sheet having passed the fuser and having reversed passes through the reconveyance path, maintain the movable chute at the second position; and after the leading edge of the sheet conveyed along the reconveyance path passes the registration roller, control the drive source to move the movable chute from the second position toward the first position.
5. The image forming apparatus according to claim 3, further comprising: a registration roller

located upstream of the transfer unit in the first direction, the registration roller being configured to correct a skew of the sheet that is conveyed toward the transfer unit, wherein the controller is configured to: after the leading edge of the sheet conveyed along the sheet conveyance path passes the fuser, control the drive source to move the movable chute from the first position toward the second position; after a trailing edge of the sheet conveyed along the sheet conveyance path passes the transfer unit, control the drive source to move the movable chute from the second position toward the first position; before a leading edge of the sheet having passed the fuser and having reversed and being conveyed along the reconveyance path reaches the first reconveyance surface, control the drive source to move the movable chute from the first position toward the second position; and after the leading edge of the sheet conveyed along the reconveyance path passes the registration roller, control the drive source to move the movable chute from the second position toward the first position.

6. The image forming apparatus according to claim 1, further comprising: a solenoid actuator configured to move the movable chute; and a chute link connected to the solenoid actuator and configured to move the movable chute from the first position to the second position, wherein the chute link is configured to be driven by the solenoid actuator to press the movable chute from a side of the sheet conveyance surface, thereby moving the movable chute from the first position to the second position.

7. The image forming apparatus according to claim 1, further comprising: a guide frame supporting the movable chute so as to be swingable, the guide frame being located upstream of the movable chute in the first direction, the guide frame having a second reconveyance surface forming the reconveyance path; and a fuser chute located upstream of the fuser and downstream of the movable chute in the first direction, the fuser chute having a third reconveyance surface forming the reconveyance path.

8. The image forming apparatus according to claim 7, wherein the fuser chute and the movable chute overlap in a sheet conveyance direction of the sheet that is conveyed along the reconveyance path.

9. The image forming apparatus according to claim 7, wherein an upstream end of the first reconveyance surface of the movable chute in a second direction is located farther from the reconveyance path than a downstream end of the third reconveyance surface of the fuser chute in the second direction, the second direction being a sheet conveyance direction of the sheet that is conveyed toward the transfer unit along the reconveyance path.

10. The image forming apparatus according to claim 7, wherein an upstream end of the second reconveyance surface of the guide frame in a second direction is located farther from the reconveyance path than a downstream end of the first reconveyance surface of the movable chute in the second direction, the second direction being a sheet conveyance direction of the sheet that is conveyed toward the transfer unit along the reconveyance path.

11. The image forming apparatus according to claim 1, wherein a downstream end of the movable chute in the first direction when the movable chute is located at the second position is located at a lower position than the downstream end of the movable chute in the first direction when the movable chute is located at the first position.

12. The image forming apparatus according to claim 11, wherein the sheet thickness direction is an upper-lower direction; and wherein the movable chute is located above the reconveyance chute, the movable chute forming an upper surface of the reconveyance path.

13. The image forming apparatus according to claim 6, wherein the chute link is configured to move the movable chute from the first position to the second position by pressing the movable chute from above.

14. The image forming apparatus according to claim 12, further comprising: a guide frame supporting the movable chute so as to be swingable, the guide frame located upstream of the movable chute in the first direction, the guide frame having a second reconveyance surface facing

the reconveyance path; and a fuser chute located upstream of the fuser and downstream of the movable chute in the first direction, the fuser chute having a third reconveyance surface facing the reconveyance path, wherein the second reconveyance surface of the guide frame and the third reconveyance surface of the fuser chute form an upper surface of the reconveyance path.

15. The image forming apparatus according to claim 14, wherein the first reconveyance surface of the movable chute and the third reconveyance surface of the fuser chute overlap as viewed from a sheet width direction, the sheet width direction crossing the first direction and the upper-lower direction.

16. The image forming apparatus according to claim 7, further comprising: a sheet metal member supported by the movable chute; and a stopper attached to the movable chute from a side of the first reconveyance surface, the stopper fixing the sheet metal member to the movable chute, wherein the stopper and the fuser chute overlap as viewed from the sheet thickness direction.

17. The image forming apparatus according to claim 1, wherein the first reconveyance surface forms the reconveyance path when the movable chute is located at the first position.

18. The image forming apparatus according to claim 17, wherein the movable chute is configured to enter the reconveyance path when the movable chute is located at the second position.

19. The image forming apparatus according to claim 1, further comprising: a fuser chute located upstream of the fuser and downstream of the movable chute in the first direction, wherein a central part of the fuser chute in a sheet width direction protrudes in a direction in which the sheet is conveyed along the reconveyance path, the sheet width direction crossing the first direction and the sheet thickness direction; and wherein the central part of the fuser chute and the movable chute overlap as viewed from the sheet thickness direction.

20. The image forming apparatus according to claim 1, further comprising: a sheet metal member supported by the movable chute, wherein the movable chute includes: a base member; and a guide rib protruding from the base member and extending in the first direction, an end surface of the guide rib forming the sheet conveyance surface; and wherein the sheet metal member is formed with a slit extending in the first direction, the guide rib being inserted through the slit, the sheet metal member being connected to ground.
